

Design FMEA

VBF-T4R2-DFMEA-01 | case t4_r2

Case t4_r2 · finalist E3-porheadroom · qualitative version based on simulation signals · 2026-08-26

Design FMEA (qualitative) — VBF-T4R2-DFMEA-01

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Qualitative S/O ratings derived from simulation-signal magnitudes vs thresholds. Rated "qualitative version, based on simulation signals"; complete FMEA (process/supplier failures) is "N/A (beyond pure simulation boundary)". No numbers from memory — signals cited from output files.

Conclusion

Highest-risk items: (1) plating margin 6 mV and (2) thermal margin 6.19 K — both PASS but are the binding margins; both mitigations are implemented in the final design (anode-porosity + transport electrolyte for plating; h=80 cooling for thermal) and are stable across two independent DFN runs. Full process/supplier-level FMEA: N/A (beyond pure simulation boundary).

Failure mode	Failure cause	Simulation signal (detectability basis)	Severity (qual.)	Occurrence (qual.)	Design-side mitigation
Negative-electrode plating during 4C fast charge	separator-interface salt depletion → anode surface potential vs Li ⁰ dips below 0 during end-of-charge (established in R2/R3 diagnostics; geometry levers insufficient)	anode_potential_v min = +0.0060 V (only 6 mV above onset); 0/307 points negative in stage-4 exam	Medium (passes, but razor-thin margin)	Low (identical in 2 independent DFN runs; levers deliberately pushed: anode porosity 0.28, t ⁺ 0.6, σ 1.6 S/m)	Keep 4C charge ≤ 0.59 Ah window as specified; do NOT raise anode porosity above 0.28 or drop electrolyte transport; if pack integration allows, re-verify with further anode margin; production control on porosity/transference targets
Thermal-runaway risk at 4C/45 °C	4C charge + pre-condition discharge heating at 45 °C ambient	T_max 326.96 K vs 333.15 (margin 6.19 K; h=80 liquid cooling required)	Medium	Low (margin stable across 2 runs; h lever measured −6.1 K per +20 units in R4)	Mandate liquid-class cooling (h ≈ 80 W/(m²·K)); derate to 3C if h < 50; monitor T during pack validation
Electrolyte oxidative decomposition	charged positive surface vs electrolyte HOMO (voltage window 4.2 V)	no true-compute HOMO/IE-EA data (Stage 5 skipped, real_compute=false); only window limit 4.2 V as signal	Medium (qualitative — not computed)	Medium (qualitative — not computed)	4.2 V limit set as designed; recommend true-compute HOMO check of electrolyte formulation before production (beyond this session's boundary)
Insufficient capacity / energy	design misses ED thresholds	margins: +132.3 Wh/kg, +56.8 Wh/L, retention +4.6 ppt (all PASS)	Low	Low	margins re-verified at DFN grade; keep compression/porosity spec
Cold-performance beyond −20 °C	rate capability below spec point	verified only at 253.15 K soak (retention 0.9938 at 1C); electrolyte σ/D are flat-in-T in this parameter set — real electrolytes worsen below −20 °C	Low at spec point	Medium beyond spec (simulation boundary — physical testing required)	qualifying below −20 °C or above 45 °C requires physical experiment (N/A in sim); noted in DVPR